Exercise - 5.5

Choose the correct or the most suitable answer.

1. The value of $2 + 4 + 6 + \cdots + 2n$ is

   (1) $\frac{n(n-1)}{2}$ (2) $\frac{n(n+1)}{2}$ (3) $\frac{2n(2n+1)}{2}$ (4) $n(n+1)$

2. The coefficient of x^6 in $(2 + 2x)^{10}$ is

   (1) ${}^{10}C_6$ (2) 2^6 (3) ${}^{10}C_6\, 2^6$ (4) ${}^{10}C_6\, 2^{10}$.

3. The coefficient of x^8y^{12} in the expansion of $(2x + 3y)^{20}$ is

   (1) 0 (2) $2^8 3^{12}$ (3) $2^8 3^{12} + 2^{12} 3^8$ (4) ${}^{20}C_8\, 2^8 3^{12}$.

4. If ${}^nC_{10} > {}^nC_r$ for all possible r, then a value of n is

   (1) 10 (2) 21 (3) 19 (4) 20.

5. If a is the arithmetic mean and g is the geometric mean of two numbers, then

   (1) $a \leq g$ (2) $a \geq g$ (3) $a = g$ (4) $a > g$.

6. If $(1 + x^2)^2 (1 + x)^n = a_0 + a_1x + a_2x^2 + \cdots + x^{n+4}$ and if a_0, a_1, a_2 are in AP, then n is

   (1) 1 (2) 5 (3) 2 (4) 4.

7. If $a, 8, b$ are in AP, $a, 4, b$ are in GP, and if a, x, b are in HP then x is

   (1) 2 (2) 1 (3) 4 (4) 16.

8. The sequence $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}+\sqrt{2}}, \frac{1}{\sqrt{3}+2\sqrt{2}}, \cdots$ form an

   (1) AP (2) GP (3) HP (4) AGP.

9. The HM of two positive numbers whose AM and GM are 16, 8 respectively is

   (1) 10 (2) 6 (3) 5 (4) 4.

10. If S_n denotes the sum of n terms of an AP whose common difference is d, the value of $S_n - 2S_{n-1} + S_{n-2}$ is

    (1) d (2) 2d (3) 4d (4) d^2.

11. The remainder when 38^{15} is divided by 13 is

    (1) 12 (2) 1 (3) 11 (4) 5.

12. The nth term of the sequence $1, 2, 4, 7, 11, \cdots$ is

    (1) $n^3 + 3n^2 + 2n$ (2) $n^3 - 3n^2 + 3n$ (3) $\frac{n(n+1)(n+2)}{3}$ (4) $\frac{n^2-n+2}{2}$.

13. The sum up to n terms of the series $\frac{1}{\sqrt{1}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{5}} + \frac{1}{\sqrt{5}+\sqrt{7}} + \cdots$ is

    (1) $\sqrt{2n+1}$ (2) $\frac{\sqrt{2n+1}}{2}$ (3) $\sqrt{2n+1} - 1$ (4) $\frac{\sqrt{2n+1}-1}{2}$.

14. The nth term of the sequence $\frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \frac{15}{16}, \cdots$ is

    (1) $2^n - n - 1$ (2) $1 - 2^{-n}$ (3) $2^{-n} + n - 1$ (4) 2^{n-1}.

15. The sum up to n terms of the series $\sqrt{2} + \sqrt{8} + \sqrt{18} + \sqrt{32} + \cdots$ is

    (1) $\frac{n(n+1)}{2}$ (2) $2n(n+1)$ (3) $\frac{n(n+1)}{\sqrt{2}}$ (4) 1.